ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

ATE MACOKE

Understanding Our Land

A Community Land Assessment Mapbook

Vegetation | Land Cover | Changes Through Time

A printed resource for discussing satellite observations alongside knowledge of the land.

Community review draft | Assessment period: 2015-2023

Reading the Land Together

This mapbook brings together satellite maps and observations developed for the Ate Macoke land assessment. It is meant to be read, discussed, and questioned.

The maps show different ways of observing vegetation and land cover. They can help us notice patterns across years and identify places that deserve a closer look.

Satellite observations are only one source of information. Familiarity with the land, seasonal conditions, past land use, and observations made on the ground are essential to interpreting what these maps mean.

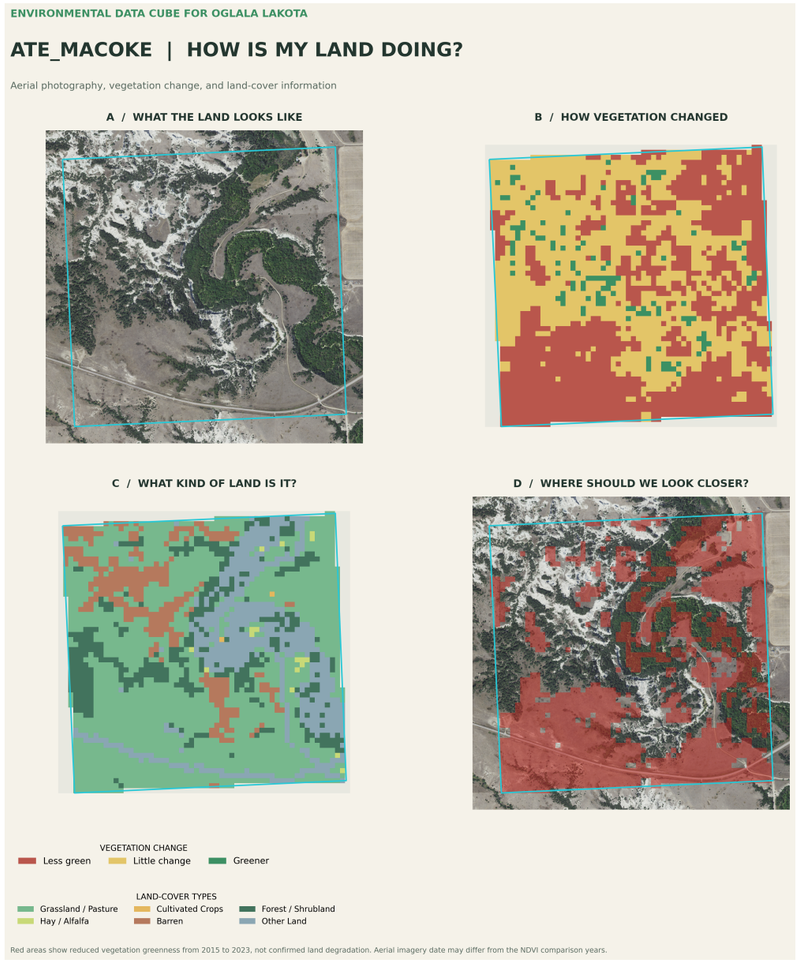
Each map includes a short explanation and a question for discussion. You do not need internet access or mapping software to use this book.

What do you notice on the land?

MAP 01 | ATE_MAP_005

How Is My Land Doing?

A first look at the assessment



WHAT THIS MAP SHOWS

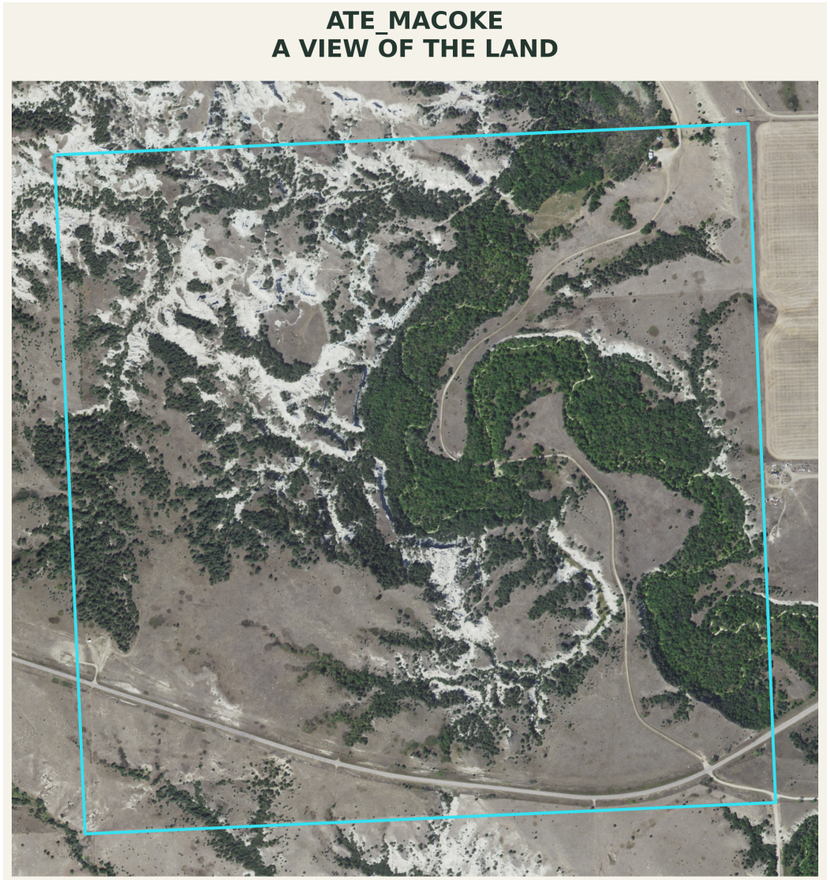
This overview brings together several ways of looking at the land. Each view tells part of the story, but no single map tells us everything.

FOR DISCUSSION: Which places would you like to look at more closely?

MAP 02 | ATE_MAP_069

A View of the Land

Seeing the parcel from above



Aerial imagery: USGS / USDA NAIP | Cyan outline: Ate_Macoke parcel

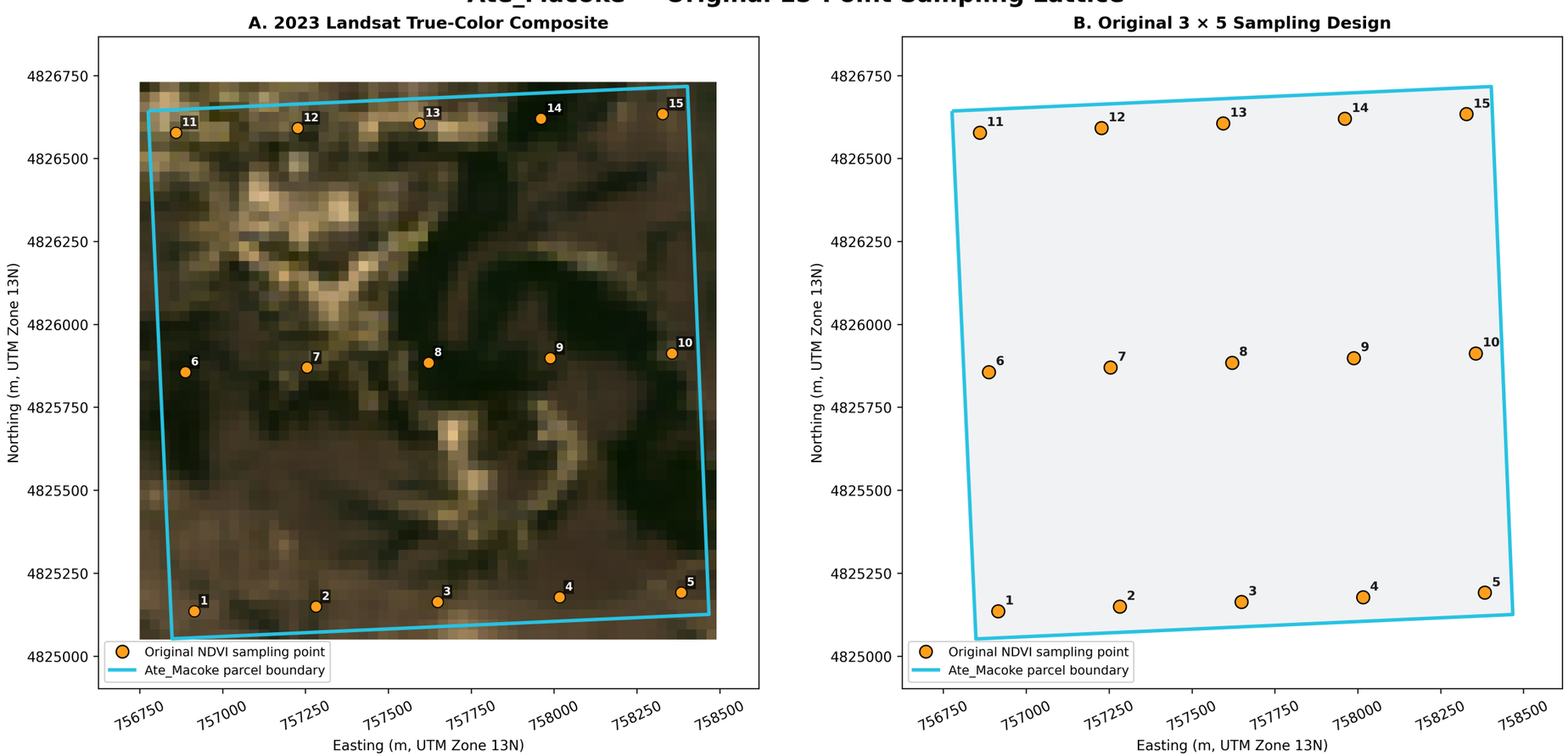
WHAT THIS MAP SHOWS

An aerial view helps us recognize the shape of the land and features visible from above. The image may have been collected in a different year than the vegetation measurements.

FOR DISCUSSION: What places or features do you recognize?

Where We Looked

The original 15-point sampling lattice



Original 15-point lattice (2015–2023) | Ate_Macoke parcel: 258.82 ha | Landsat 2023 RGB: 30 m

WHAT THIS MAP SHOWS

These sampling locations show where vegetation observations were organized for part of the analysis. They are measurement locations, not proof of conditions everywhere on the parcel.

FOR DISCUSSION: Are there other places that should be examined?

Vegetation Through Time

Looking across the assessment years

WHAT THIS MAP SHOWS

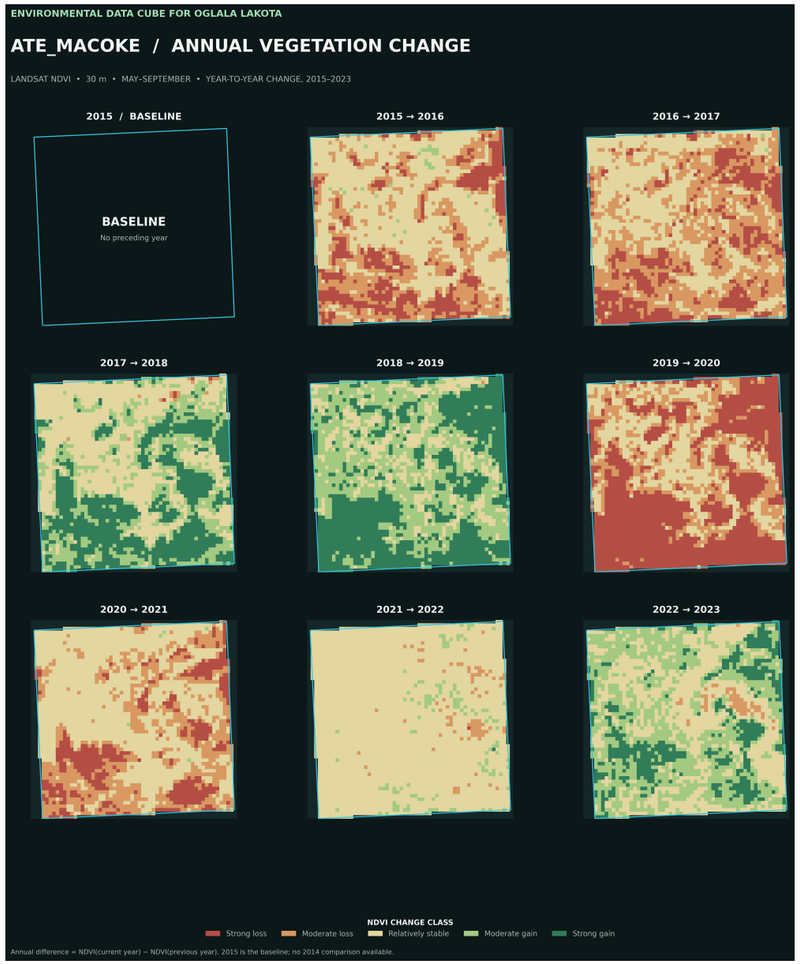
Satellite vegetation measurements help us compare greenness over time. Seasonal weather, rainfall, grazing, and other factors can all affect these patterns.

FOR DISCUSSION: Which years seem different from what you remember?

MAP 05 | ATE_MAP_073

How Vegetation Changed

Changes across the observation period



WHAT THIS MAP SHOWS

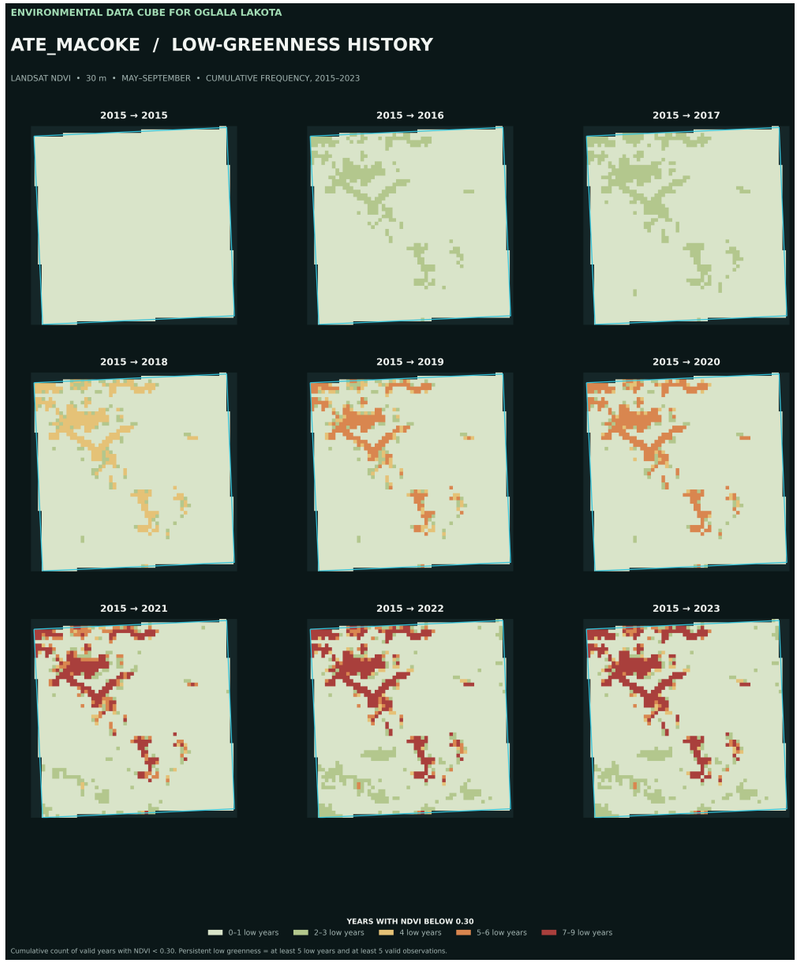
Changes in vegetation greenness can help us identify patterns worth discussing. A decrease in greenness does not, by itself, establish overgrazing or land degradation.

FOR DISCUSSION: Where do the changes match your experience?

MAP 06 | ATE_MAP_074

Places With Lower Greenness

Looking for recurring patterns



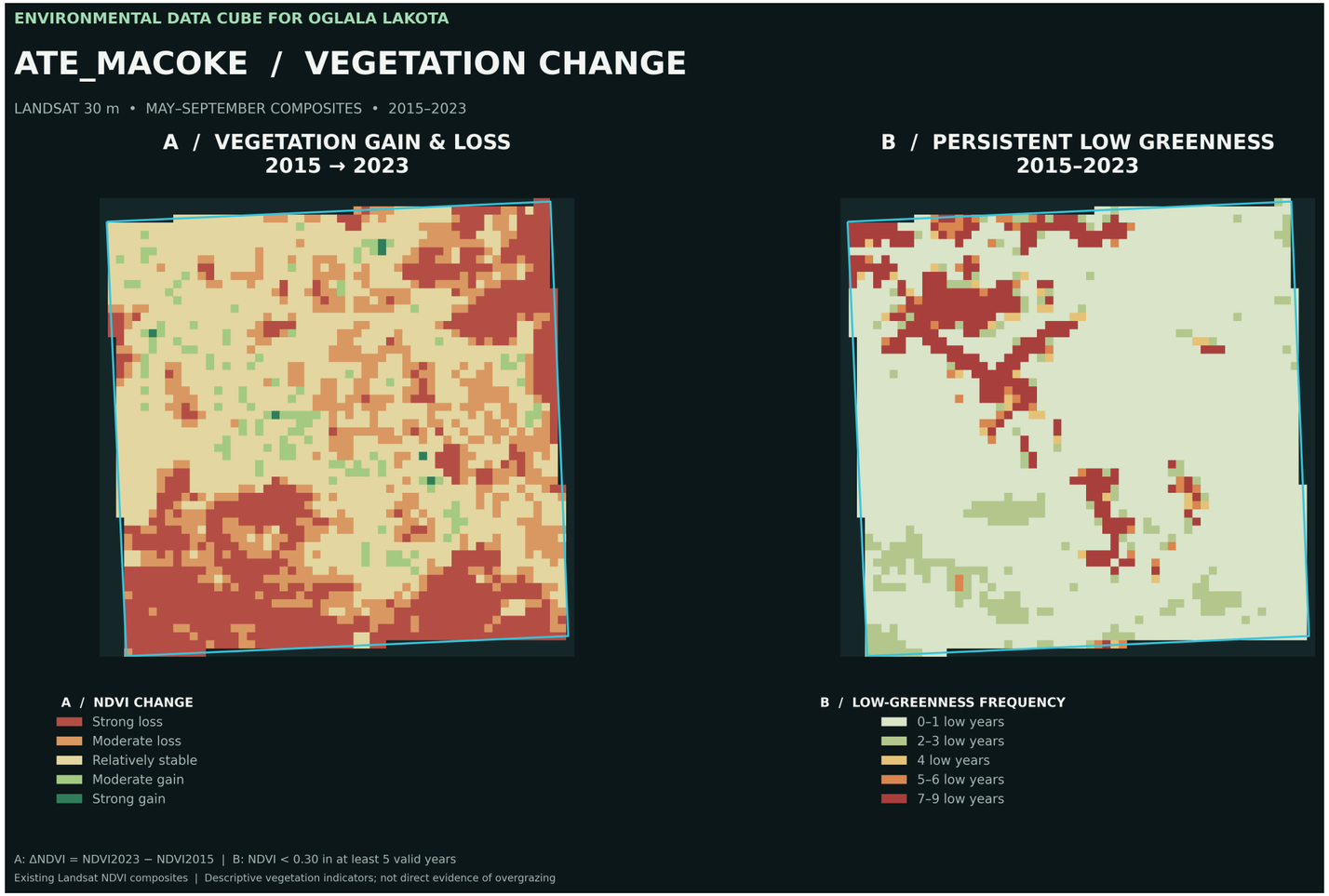
WHAT THIS MAP SHOWS

This map highlights patterns of lower greenness across observations. These patterns may reflect several causes and need local knowledge or field observations to understand.

FOR DISCUSSION: Are these places naturally different, or have you noticed changes there?

Change and Persistence

Understanding different vegetation patterns



WHAT THIS MAP SHOWS

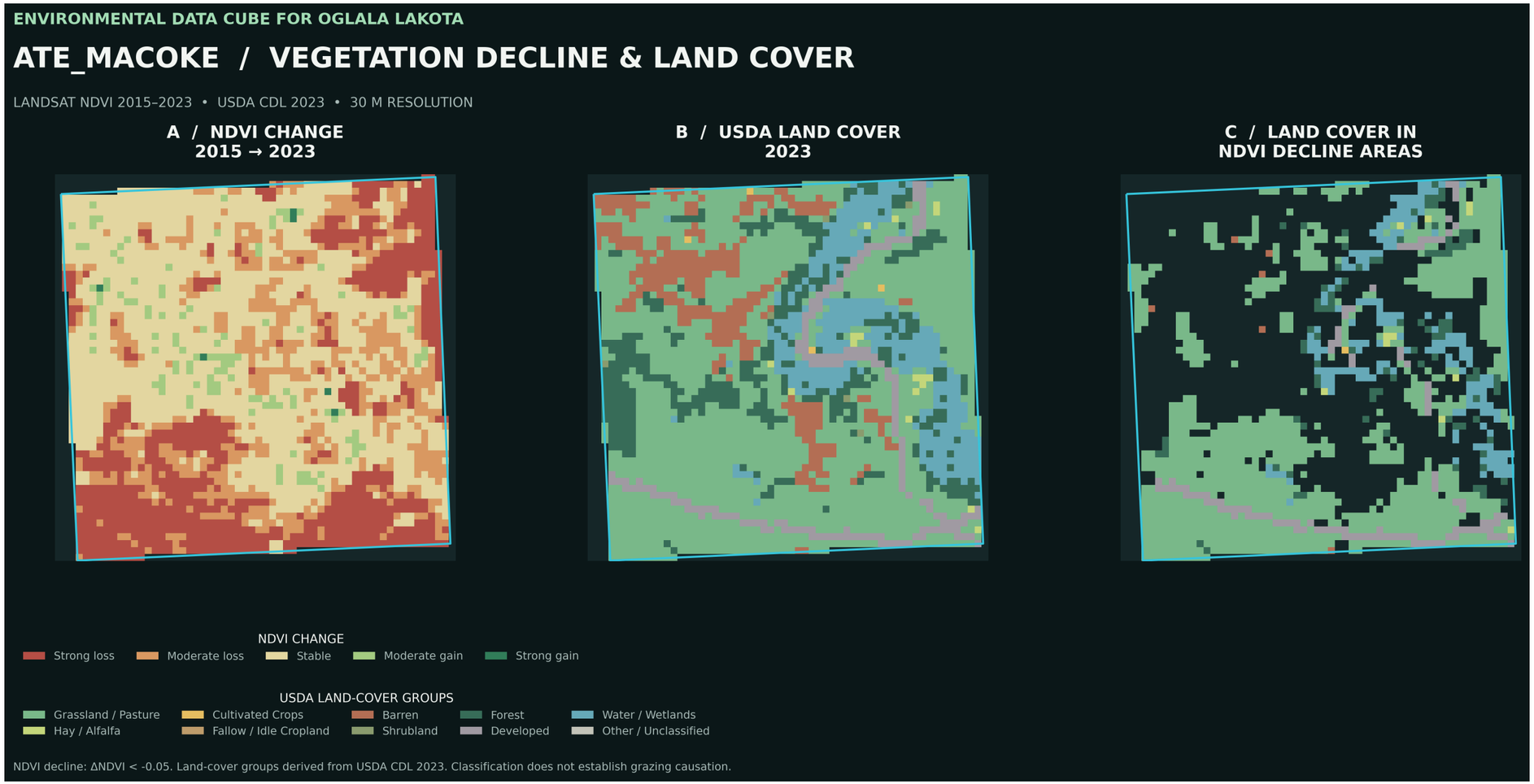
Some places may change over time while others show more consistent patterns. Comparing these patterns can help guide conversations about land conditions.

FOR DISCUSSION: Which areas would be useful to revisit together?

MAP 08 | ATE_MAP_011

Where Should We Look Closer?

Vegetation and land-cover observations



WHAT THIS MAP SHOWS

This map helps identify candidate areas for closer attention. They are places to ask questions, not confirmed problem areas.

FOR DISCUSSION: Where would you begin a visit to the land?

What These Maps Can Tell Us

Vegetation greenness is a useful satellite indicator, but it is not a direct measurement of grazing pressure, soil health, or overall land condition.

A place showing less greenness may have been affected by drought, seasonal differences, plant communities, land use, or other factors. Lower greenness alone does not confirm overgrazing or land degradation.

Different maps may use imagery collected on different dates. Aerial photographs and vegetation measurements should not be assumed to represent the same day or year.

Map boundaries, classification labels, measurement dates, and field conditions should be checked before making management decisions.

These maps are a starting point for conversation and closer observation, not a final judgment about the land.

Observations From the Land

Use this page to record places, seasons, changes, questions, or observations that help explain what the maps show.

Places I recognize:

Changes I have noticed:

Things the maps may not show:

Places to visit or examine: